Procedures in MySQL – Exercise 16

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Program 1 – Procedure to get all the records from the table student_info

1. CREATE A DATABASE AND TABLE

   CREATE DATABASE IF NOT EXISTS CollegeDB;
   USE CollegeDB;
   CREATE TABLE student_info (RollNo INT PRIMARY KEY, Name VARCHAR(50), Department VARCHAR(20), Marks INT);

   Output:

```
Query OK, 1 row affected (0.048 sec)

Database changed

Query OK, 0 rows affected (0.235 sec)
```

2. INSERT RECORDS

   INSERT INTO student_info VALUES
   (101, 'Arun', 'CSE', 85),
   (102, 'Bala', 'IT', 90),
   (103, 'Charan', 'ECE', 78),
   (104, 'Divya', 'CSE', 92),
   (105, 'Eswar', 'MECH', 69);

   Output:

```
Query OK, 5 rows affected (0.045 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

3. CREATE THE PROCEDURE

```
DELIMITER //
CREATE PROCEDURE GetStudentInfo()
BEGIN
    SELECT * FROM student_info;
END //
DELIMITER ;
```

   Output:

```
Query OK, 0 rows affected (0.039 sec)
```

4. CALL THE PROCEDURE
   CALL GetStudentInfo();

   Output:

```
+--------+--------+------------+-------+
| RollNo | Name   | Department | Marks |
+--------+--------+------------+-------+
|    101 | Arun   | CSE        |    85 |
|    102 | Bala   | IT         |    90 |
|    103 | Charan | ECE        |    78 |
|    104 | Divya  | CSE        |    92 |
|    105 | Eswar  | MECH       |    69 |
+--------+--------+------------+-------+
5 rows in set (0.004 sec)

Query OK, 0 rows affected (0.005 sec)
```

Program 2 – Procedure GetCustomerLevel() with IN and OUT parameters

5. CREATE THE CUSTOMERS TABLE AND INSERT RECORDS
   CREATE TABLE customers (customerNumber INT PRIMARY KEY, customerName VARCHAR(50), creditLimit DECIMAL(10,2));

   INSERT INTO customers VALUES
   (101, 'Kumar Traders', 21000.00),
   (102, 'Sri Gifts', 71800.00),
   (103, 'Bala Stores', 0.00),
   (104, 'Lakshmi Mart', 9500.00),
   (105, 'Anand and Co', 50000.00);

   Output:

```
Query OK, 0 rows affected (0.173 sec)

Query OK, 5 rows affected (0.053 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

6. VIEW THE CUSTOMERS TABLE

   SELECT * FROM customers;

   Output:

```
+----------------+---------------+-------------+
| customerNumber | customerName  | creditLimit |
+----------------+---------------+-------------+
|            101 | Kumar Traders |    21000.00 |
|            102 | Sri Gifts     |    71800.00 |
|            103 | Bala Stores   |        0.00 |
|            104 | Lakshmi Mart  |     9500.00 |
|            105 | Anand and Co  |    50000.00 |
+----------------+---------------+-------------+
5 rows in set (0.003 sec)
```

7. CREATE THE PROCEDURE GetCustomerLevel

```
DELIMITER $$
CREATE PROCEDURE GetCustomerLevel(
    IN  p_customerNumber INT,
    OUT p_customerLevel  VARCHAR(20)
)
BEGIN
    DECLARE creditlim DECIMAL(10,2);

    SELECT creditLimit INTO creditlim
    FROM customers
    WHERE customerNumber = p_customerNumber;

    IF creditlim > 50000 THEN
        SET p_customerLevel = 'PLATINUM';
    ELSEIF creditlim >= 10000 THEN
        SET p_customerLevel = 'GOLD';
    ELSE
        SET p_customerLevel = 'SILVER';
    END IF;
END$$
DELIMITER ;
```

   Output:

```
Query OK, 0 rows affected (0.030 sec)
```

8. CALL THE PROCEDURE FOR CUSTOMER 101

   CALL GetCustomerLevel(101, @level);

   SELECT @level AS customerLevel;

   Output:

```
Query OK, 0 rows affected (0.002 sec)

+---------------+
| customerLevel |
+---------------+
| GOLD          |
+---------------+
1 row in set (0.000 sec)
```

9. CALL THE PROCEDURE FOR CUSTOMER 102

```
CALL GetCustomerLevel(102, @level);
SELECT @level AS customerLevel;
```

Output:

```
Query OK, 0 rows affected (0.004 sec)

+---------------+
| customerLevel |
+---------------+
| PLATINUM      |
+---------------+
1 row in set (0.000 sec)
```

10. CALL THE PROCEDURE FOR CUSTOMER 103

```
CALL GetCustomerLevel(103, @level);
SELECT @level AS customerLevel;
```

Output:

```
Query OK, 0 rows affected (0.002 sec)

+---------------+
| customerLevel |
+---------------+
| SILVER        |
+---------------+
1 row in set (0.000 sec)
```

11. CALL THE PROCEDURE FOR CUSTOMER 105

```
CALL GetCustomerLevel(105, @level);
SELECT @level AS customerLevel;
```

Output:

```
Query OK, 0 rows affected (0.005 sec)

+---------------+
| customerLevel |
+---------------+
| GOLD          |
+---------------+
1 row in set (0.000 sec)
```